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Lower Jurassic palaeomagnetic results from Yorkshire, England, and their implications

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Palaeomagnetic study of Middle Liassic siltstones shows a stable magnetization with a mean direction of $D = 12.3°$, $I = 64.6°$ ($N = 60$, $k = 26$, $\alpha_{95} = 3.9°$) corresponding to a palaeomagnetic pole at 79.8°N, 125.6°E, similar to that for southern Germany and confirming predictions based on palaeogeographic reconstructions using North American data. Sideritic concretions of Lower Liassic age show a higher magnetic stability with a mean direction of $D = 12.6°$, $I = 61.4°$ ($N = 125$, $k = 50$, $\alpha_{95} = 1.8°$) which is not significantly different from the siltstones. This confirms the sedimentological evidence that suggests that such concretions grew very shortly after deposition, i.e. within the Liassic, and suggests that similar concretions of other ages could thus be used for palaeomagnetic studies. Although the Liassic palaeomagnetic pole (76.9°N, 134.7°E), based on this work, appears valid it is still not possible to evaluate a sensible Mesozoic polar wandering curve for the North Atlantic bordering continents.

1. Introduction

European and North American palaeomagnetic pole determinations for the Jurassic are particularly significant as they provide palaeogeographical constraints on these areas during the early formation of the North Atlantic and the deposition of oil shales on their edges. The palaeomagnetic record for both continents is sparse and generally of poor reliability (see discussion). In order to improve the data available, and to study the mode of formation of calcareous nodules, Lower Jurassic samples were collected from well exposed sites near Staithes, Yorkshire (54.6°N, 0.8°W). These comprise the Lower Pliensbachian Ironstone Shales (dark grey to pale grey micaceous and silty shales with common sideritic mudstone concretions) and the Middle Pliensbachian Staithes Formation (fine- to medium-grained, carbonate-cemented, rnicaceous sandstones, with lenses of sideritic mudstone). At least six separately oriented cores or hand samples were collected from each of 45 sites in the two formations, and one or two specimens were cut from each core. All mesaurements were made with a Digico spinner magnetometer and were subjected to thermal demagnetization using standard techniques including the monitoring of low-field susceptibility as a check on the occurrence of chemical changes during heating.

2. Palaeomagnetic studies

The intensity of the initial natural remanent magnetization (NRM) of all the specimens varied from 0.02 to 11.0 $mA\ m^{-1}$. The mean site initial NRM directions were closely grouped (Fig. la) and no statistical difference was found between the corrected and uncorrected NRM directions, mainly because the bedding dips were low and fairly uniform. Thermal demagnetization was applied to 40 pilot specimens by heating to 50°C and then in steps of 50°C up to the peak temperature

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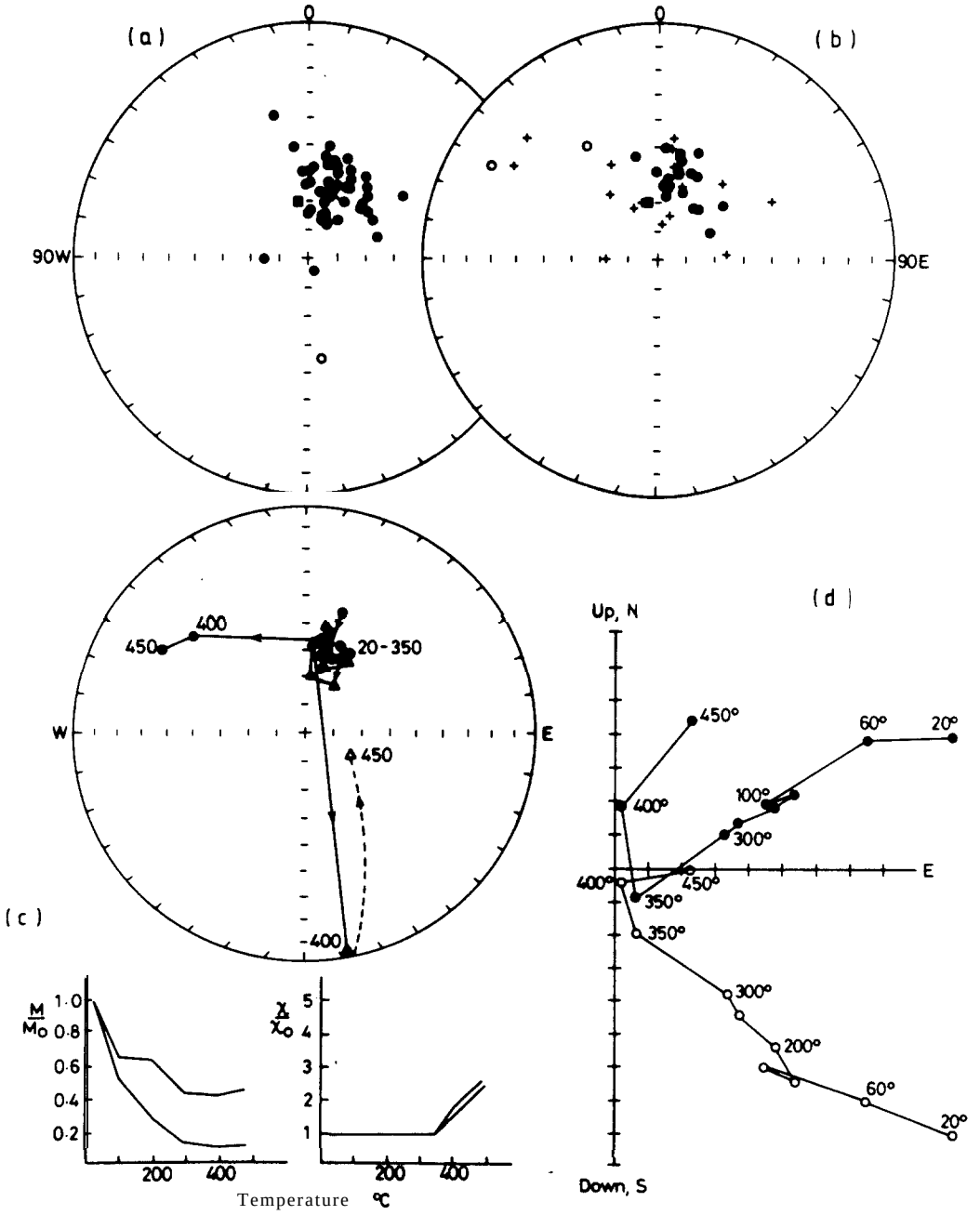


Fig. 1. Stereographic projections of the site mean directions: (a) the initial Staithes site mean magnetic directions; (b) the magnetic directions over the most stable range; (c) the change in magnetic direction, for typical samples (ST 73.2 and ST 84.4), along with their appropriate normalized susceptibility and intensity curves; and (d) the Zijderveld diagram showing one main component up to 350°C. In (a) the full circles correspond to downward inclination, open circle corresponds to upward inclination, solid square corresponds to Earth's present magnetic direction, and cross symbol corresponds to overall mean direction. Full circle symbols in (b) correspond to stable downward magnetic direction while open circle and plus symbols correspond to unstable upward and downward magnetic directions respectively. Full and open circle symbols in (d) corresponds to x y and y z components, respectively.

above which either the intensity became too weak for adequate measurement (0.02 $mA\ m^{-1}$) or major chemical changes had taken place, as indicated by major low-field susceptibility changes. The changes in magnetic direction, along with appropriate normalized intensity and susceptibility curves, indicate two fundamental types of behaviour during thermal demagnetization:

(1) Group I, comprising 23 of the 40 pilot specimens, showed similar decreases in intensity and consistent changes in susceptibility (Fig. lc, d). All these samples were from either the Staithes

TABLE 1

Cleaned site mean data: Staithes Lower Jurassic

Site	N_{sp}	D	I	α_{95}	k
Siltstone sites (Stoithes Formation) [a]					
ST 65	8	10.6	54.4	6.8	68
ST 71	7	12.7	61.8	12.0	26
ST 77	6	9.0	67.6	10. 1	45
ST 78	5	14.4	63.1	1.1	99
ST 79	5	28.1	68.7	11.1	48
ST 80	6	17.8	57.1	16.0	18
ST 81	6	34.1	69.9	15.1	21
ST 83	6	345.0	71.8	13.2	27
ST 84	6	349.3	59.8	11.6	16
ST 85	6	8.4	65.2	33.1	5
ST 86	5	27.7	61.8	16.6	22
Mean	60	12.3	64.6	3.9	26
Concretions (Ironstone Shales) [b]					
ST 21	8	9.2	54.3	1.9	50
ST 22	8	9.6	55.7	4.6	144
ST 23	8	3.8	58.4	5.6	100
ST 60	7	351.4	57.6	4.8	156
ST61	8	354.9	60.0	3.7	219
ST 63	6	42.5	70.7	4.0	277
ST 64	7	354.4	59.0	3.9	239
ST 66	7	5.6	56.1	6.8	80
ST 67	4	3.9	54.8	9.0	106
ST 69	7	359.9	64.8	4.6	171
ST 70	6	2.6	59.7	3.3	420
ST 70B	6	39.6	56.8	5.9	129
ST 71B	6	20.1	65.1	32.1	7
ST 72	7	15.9	59.2	6.9	77
ST 73	7	12.2	63.7	5.5	120
ST 75	6	24.8	62.7	10.5	42
ST 74	6	37	68.7	10.3	43
ST 76	6	39.7	65.8	8.7	61
ST 88	6	33.0	60.7	5.4	155
ST 89	6	22.2	64.2	5.9	169
Mean	125	12.6	61.4	1.8	50
Overal mean [c]	185	12.4	62.3	1.7	37

[a] Site ST 85 is omitted from the mean; palaeomagnetic pole = 79.8°N, 125.6°E ($N = 60$, $k = 13$, $\alpha_{95} = 5.6°$).
[b] Site ST 71B is omitted from the mean; palaeomagnetic pole = 76.1°N, 139.6°E ($N = 125$; $k = 25$; $\alpha_{95} = 2.7°$).
[c] Paiaeomagnetic pole = 76.9°N, 134.7°E ($N = 185$, $k = 19$, $\alpha_{95} = 2.5°$).

Formation or from nodules within the Ironstone Shales. Twenty-one of these 23 pilot specimens also showed good magnetic stability, i.e. stability indices [I] greater than 2.5, and only the two poorly stable specimens (ST 31.4 and 82.3) were excluded from subsequent analyses. The mean direction of the 21 stable specimens, after correction for dip is 16.6' + 61.7' (k = 69.0, α_{95} = 3.8"). This is close to, but statistically different from, both the present geomagnetic field direction at the locality (350°, +69.5°) and from the local axial geocentric dipole field direction (0°, + 70.4').

(2) Group II did not show common intensity or susceptibility behaviour and, during demagnetization, all 17 pilot specimens showed unstable to poorly stable behaviour, except for ST 39.6, 40.1 and 48.2. These three exceptional specimens had stable to very stable properties and had been deliberately collected from weathered outcrops to test for the effects of recent weathering. None of the group II samples are considered representative of the Jurassic magnetic field direction and are not considered further in this context.

All the remaining specimens from 31 group I sites were then partially demagnetized by heating to 200°C, being the optimum temperature at which the most stable direction appeared to have been isolated in studies of their pilot specimens. This cleaning resulted in similar site mean directions as had been obtained from studies of their individual specimens, with the siltstone showing a somewhat larger within- and between-site scatter than for the concretions (Table 1). In addition, the unweathered specimens were examined from sites in which the stably magnetized weathered remanence had been obtained. The "unweathered" specimens showed similar instability as observed in other group II samples while the weathered samples, after heating to 150–250°C, had a mean direction (345.0°, 66.9"; α_{95} = 15.3) close to that of the present geomagnetic field direction (350°, 69.5°). The weathered samples also showed a much higher intensity of remanence (0.2-l 1 mA m^{-1}) than the unweathered samples (0.02-0.4 mA m^{-1}).

3. Interpretation, mineralogy and discussion

3.1. Concretions

Two thirds of the samples used in this study were from concretions. In the Ironstone Shales these usually occur in "rows" along bedding planes. They are dominantly sideritic in composition [2] with some ankerite [3] and an admixture of clay, silt and chlorite, i.e. the constituent minerals of the country rock. The individual concretions are not always of uniform composition throughout, for example a concretion with a completely calcitic core may become increasingly sideritic towards the surface [2]. The lateral extent of some individual beds of concretions means that the factors controlling their growth were equally extensive [4]. The concretions often provided excellent preservation conditions within them for fossils and so the formation of nodules and preservation of fossils are widely regarded as genetically connected [2]. Raiswell [5] pointed out that the concretions may be dated relative to each other and, according to his classification, the concretions in the Ironstone Shales appear to conform to the characteristics of his type 1 lb, i.e. they grew during an early period of negligible deposition and thus their remanence is likely to have been acquired very soon after the deposition of the enclosing sediments. However, the initial remanent directions of specimens, taken from the outermost part of the concretions, were slightly more scattered than those taken from within the core. This difference is not statistically significant at a 95% level, but could still indicate that there is a change in the conditions during the growth of the concretion [6]. On this basis all the specimens used in subsequent analysis were taken from the inner parts of the concretions.

If it is presumed that these concretions acquired their magnetization prior to the compaction of their upper surface, then this compaction should cause a distortion of the magnetic directions, while if they acquired their remanence after compaction then they should show consistency in direction. On this basis, a detailed study was made of two concretions (ST 88 and 89) in order to determine if their magnetizations were acquired prior to the curvature of their upper surfaces. Six cores were

drilled in the field from each of the two concretions (Fig. 2). Measurements of dip and strike of the upper surface of the concretion of each position were taken before drilling. Table 2 shows the mean initial NRM directions, with their statistical parameters both before and after the upper surface curvature corrections. The level of significance of the "fold test" is positive at greater than 99% [7] and mean that both concretions acquired their magnetization post-depositionally, i.e. most probably during their growth by radial displacement of calcite during compaction. On the other hand all concretions showed the same mean magnetic declination as the Staithes Formation siltstones (Middle Pliensbachian) and only a slight difference, 3.2°, in their mean inclinations (Table 1). Such a difference, which is not statistically significant, could represent a short time gap between them so that the concretions most probably acquired their remanence during the Pliensbachian, possibly even during the Lower Pliensbachian.

Isothermal remanence of both unheated and heated (400°C) specimens of both concretions and siltstones show the presence of magnetite (Fig. 3). Strong chemical changes in both types of specimen at slightly higher temperatures are associated with large increases in both the low-field susceptibility and saturation moment, indicating that these chemical changes are related to the formation of additional or maghemite. It is generally considered that where siderite is abundant, as in these rocks [2], not much pyrite can form [8,9] and so it seems most probable that it is magnetite that formed at 400–450°C derived by the high-temperature oxidation of siderite [10] rather than the breakdown of pyrite. Maghemite formation, which can

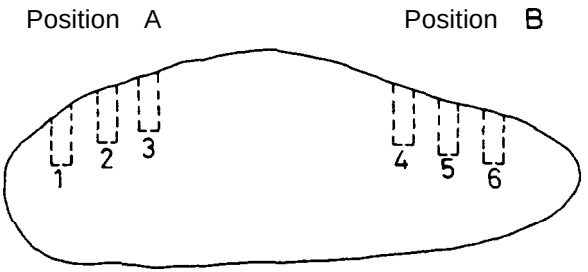


Fig. 2. The position of the six cores drilled from each of the concretions.

take place at similar temperatures, but this inverts rapidly to haematite, and it is not until after cooling from temperatures above 530°C that the isothermal remanence curves of both types show only the presence of haematite. It thus seems probable that the siderite decomposes above 400°C to form magnetite and, with pre-existing magnitites, almost immediately begin to oxidize to haematite but this is not complete until 530°C. This would also suggest that the original magnetite in these beds originated from the low-temperature, early diagenetic oxidation of siderite and its remanence is thus of chemical rather than depositional origin.

The behaviour of the concretions in this study suggest that similar nodules, irrespective of age, are likely to yield usable palaeomagnetic properties that can be related to very early stage diagenetic changes, and thus can be generally considered to broadly correspond in age to the geological age of the beds within they are found. It is also evident that the degree of chemical change of temperatures above 400°C will cause the destruction of the original diagenetic grains and their associated magnetization.

TABLE 2

The mean initial NRM of the two sites (ST 88 and 89) before and after the upper surface curvature

Site	N	Before correction			After correction			k_1/k_2	Level of significance
		D	I	k_1	D	I	k_2		
ST88	6	35.2	15.6	128	143.8	73.1	26	4.92	greater than 99% level
ST 89	6	4.1	71.8	38.0	353.8	54.7	5.0	7.6	greater than 99% level

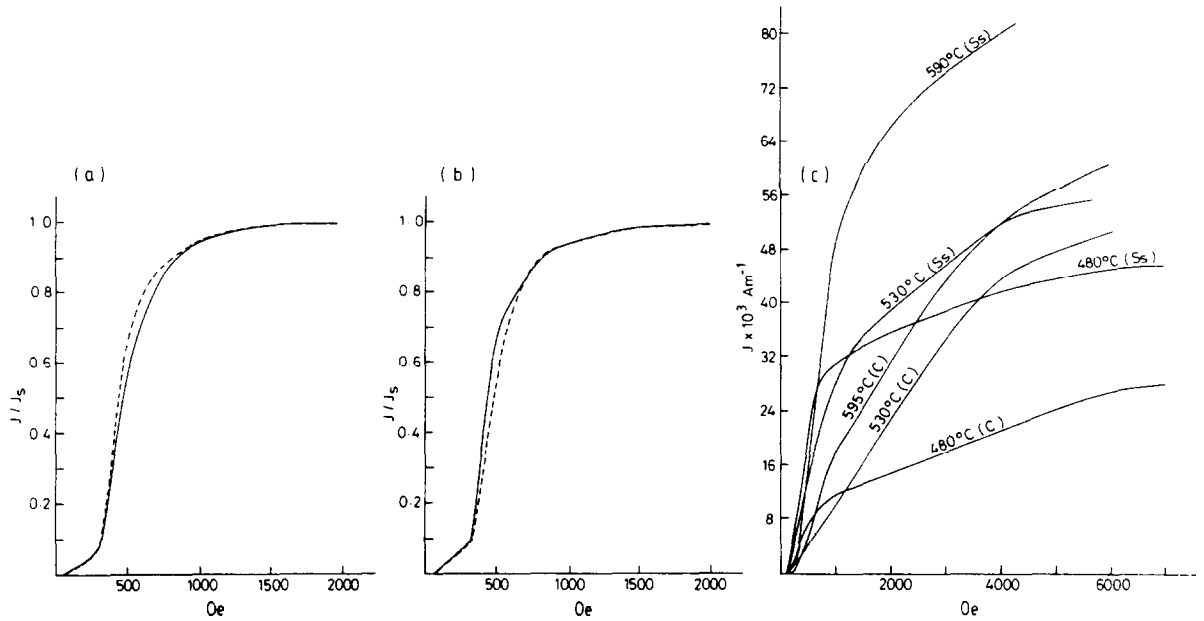


Fig. 3. Examples of the IRM curves for the heated and unheated specimens of siltstone (Staithes Formation) and concretion (Ironstone Shales). Solid and broken lines in (a) and (b) correspond to unheated and heated (400°C) specimens, respectively. *Ss* and C in (c) correspond to siltstone and concretion.

3.2. The Jurassic pole position

European Jurassic pole position. Although Jurassic rocks have been quite extensively studied in Europe, almost all observations have been universally regarded as indicating instability and poor definition of even the possible polarity of the geomagnetic field during Jurassic times. In fact, only one pole position (pole *1*, Fig. 4) can be considered to be fully reliable, based on studies of an Upper Jurassic limestone in southern Germany [11], although two other positions can be regarded as indicative of its location, in addition to the new observations. These few pole positions (Fig. 4) cannot, however, be regarded as definitive. Two of the poles (2 and 3, Fig. 4) are from igneous rocks of uncertain age, i.e. Spitsbergen diabase [12] of Upper Jurassic to Lower Cretaceous age, comprising only 8 samples, and Upper Triassic to Lower Jurassic volcanics from the northern Pyrenean foothills [13]. The latter may thus also have been tectonically disturbed and may not be representative of tectonically stable Europe. By far the most data available are from southern Russia, 19 pole

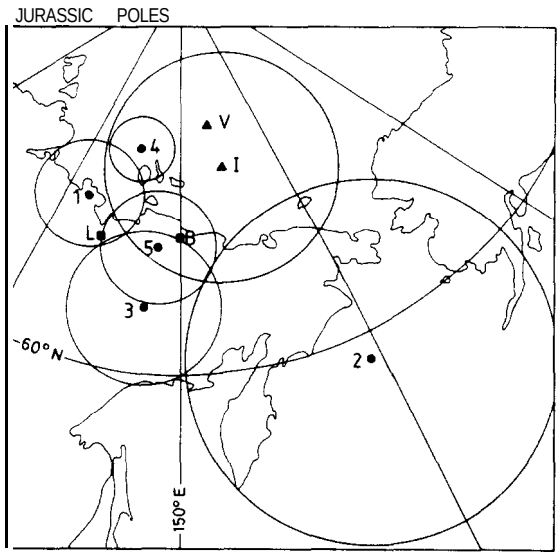


Fig. 4. The Jurassic pole positions for Western Europe. For the poles I, 2, 3, 4, and 5 see the text. *V* and I corresponds to Van der Voo and French [15] and Irving [16], respectively. *B* and *L* corresponds to the North American mean Jurassic pole after the rotation to the Bullard et al. [17] fit of North Atlantic and the Le Pichon et al. [18] fit, respectively.

positions [14], mostly for igneous rocks in very close proximity to each other and in an area very likely to have been affected by later orogenic disturbances. The techniques used in their analysis also differ from standard "western" methods, but the mean pole is quite well defined (pole 5, Fig. 4) and also lies close to the southern German pole (1 in Fig. 4).

To attempt to assess the problematic pole position for Europe in both Cretaceous and Jurassic times, continental reconstructions have been used to transfer North American pole positions to Europe where they can either be combined with the European data [15] to obtain a mean (pole V, Fig. 4), or simply using the North American mean pole [16] after rotation (pole I, Fig. 4), even though the North American Jurassic pole position is itself not well defined and the actual rotations used may be those proposed by Bullard et al. [17] or by Le Pichon et al. [18] — poles B and L, respectively, in Fig. 4.

The new Liassic pole position, based on the Staithes Formation (pole 4, Fig. 4) is not dissimilar to that found for Liassic age sediments from southern Britain (work in progress, E.A. Hailwood and C.M. Brown, personal communication) and is close to the mean American pole position, after either rotation as well as being similar to that for southern Germany and southern Russia. Nonetheless, the location of the European palaeomagnetic pole for the Jurassic period is only becoming slightly more closely defined.

4. Polarity

The normal polarity in the Staithes specimens is consistent with a normal "Jurassic Quiet Zone" [19] during the Pliensbachian, but mixed geomagnetic polarity has been reported for the Lower Jurassic rocks by some authors [20–22]. However, there are large uncertainties in their ages and even in their characteristic remanence, so at this stage, no polarity correlations can be considered valid until much more reliable Jurassic palaeomagnetic data are available.

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